

Technical Documentation: Ollama API Infrastructure

1. Infrastructure Overview

The infrastructure is deployed on Microsoft Azure using a Virtual Machine tailored for the Ollama API service. The configuration ensures a balance between performance and cost-efficiency for hosting Large Language Models (LLMs).

Virtual Machine: Standard D2s v3 (2 vCPUs, 8 GiB RAM).

Operating System: Ubuntu Server 24.04 LTS - Gen2.

Region: France Central.

Network Security: SSH port (22) and API port (11434) are configured for access, with trusted launch security enabled.

Azure vm creation :

Price

1 X Standard D2s v3

by Microsoft

[Terms of use](#) [Privacy policy](#)

Subscription credits apply

0.1120 USD/hr

[Pricing for other VM sizes](#)

You have set SSH port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

Basics

Subscription

Azure for Students

Resource group

hackathon

Virtual machine name

techcorp-ollama-api

Region

France Central

Availability options

Availability zone

Zone options

Azure-selected zone (Preview)

Security type
Trusted launch virtual machines
Enable secure boot
Yes
Enable vTPM
Yes
Integrity monitoring
No
Image
Ubuntu Server 24.04 LTS - Gen2
VM architecture
x64
Size
Standard D2s v3 (2 vcpus, 8 GiB memory)
Enable Hibernation
No
Authentication type
SSH public key
Username
azureuser
SSH Key format
Ed25519
Key pair name
techcorp-ollama-api_key
Public inbound ports
SSH
Azure Spot
No

Disks

OS disk size
Image default
OS disk type
Standard HDD LRS
Use managed disks
Yes
Delete OS disk with VM
Enabled
Ephemeral OS disk
No

Networking

Virtual network

(new) techcorp-ollama-api-vnet

Subnet

(new) default (10.0.0.0/24)

Public IP

(new) techcorp-ollama-api-ip

Accelerated networking

Off

Place this virtual machine behind an existing load balancing solution?

No

Delete public IP and NIC when VM is deleted

Enabled

Management

Microsoft Defender for Cloud

Basic (free)

System assigned managed identity

Off

Login with Microsoft Entra ID

Off

Auto-shutdown

Off

Backup

Disabled

Enable periodic assessment

Off

Enable hotpatch

Off

Patch orchestration options

Image Default

Monitoring

Alerts

Off

Boot diagnostics

Off

Enable OS guest diagnostics

Off

Enable application health monitoring

Off

Advanced

Extensions

None

VM applications

None

Cloud init

No

User data

No

Disk controller type

SCSI

Proximity placement group

None

Capacity reservation group

None

 Delete  Cancel  Redeploy  Download  Refresh

Your deployment is complete



Deployment name : CreateVm-canonical.ubuntu-24_04-lts-server-20260701105151

Subscription : [Azure for Students](#)

Resource group : [hackathon](#)

 **Deployment details**

 **Next steps**

[Set up auto-shutdown](#) Recommended

[Monitor VM health, performance, and network dependencies](#) Recommended

[Run a script inside the virtual machine](#) Recommended

[Go to resource](#)

[Create another VM](#)

[Scale out your VM](#)





Add inbound security rule



techcorp-ollama-api-nsg

Source ⓘ

Any



Source port ranges * ⓘ

*

Destination ⓘ

Any



Service ⓘ

Custom



Destination port ranges * ⓘ

11434



Protocol



Any



TCP



UDP



ICMPv4



ICMPv6

Action



Allow



Deny

Priority * ⓘ

310



Name *

Allow-Ollama-API



Description

```
System load: 0.08          Processes:           121
Usage of /:  5.8% of 28.02GB Users logged in:       0
Memory usage: 3%          IPv4 address for eth0: 10.0.0.4
Swap usage:  0%
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

```
azureuser@techcorp-ollama-api:~$ |
```

2. Ollama Installation

The Ollama service is installed via the official automated script, which handles the binary deployment and initial service setup.

Shell

```
curl -fsSL https://ollama.com/install.sh | sh
```

```
azureuser@techcorp-ollama-api:~$ curl -fsSL https://ollama.com/install.sh | sh
>>> Installing ollama to /usr/local
>>> Downloading ollama-linux-amd64.tar.zst
##### 100.0%
>>> Creating ollama user...
>>> Adding ollama user to render group...
>>> Adding ollama user to video group...
>>> Adding current user to ollama group...
>>> Creating ollama systemd service...
>>> Enabling and starting ollama service...
Created symlink /etc/systemd/system/default.target.wants/ollama.service → /etc/systemd/system/ollama.service.
>>> The Ollama API is now available at 127.0.0.1:11434.
>>> Install complete. Run "ollama" from the command line.
WARNING: No NVIDIA/AMD GPU detected. Ollama will run in CPU-only mode.
azureuser@techcorp-ollama-api:~$ |
```

3. Ollama Configuration for External Access

By default, Ollama listens only on localhost. To allow the API to be accessible from external sources, a systemd override is required to set the environment variables.

Create the override directory: `sudo mkdir -p /etc/systemd/system/ollama.service.d`

Edit the override file: Create `/etc/systemd/system/ollama.service.d/override.conf` with the following content:

None

[Service]

Environment="OLLAMA_HOST=0.0.0.0"

Reload and Restart:

Shell

`sudo systemctl daemon-reload`

`sudo systemctl restart ollama`

```
### Editing /etc/systemd/system/ollama.service.d/override.conf
### Anything between here and the comment below will become the contents of the drop-in file

### Edits below this comment will be discarded

### /etc/systemd/system/ollama.service
# [Unit]
# Description=Ollama Service
# After=network-online.target
#
# [Service]
Environment="OLLAMA_HOST=0.0.0.0"
# ExecStart=/usr/local/bin/ollama serve
# User=ollama
# Group=ollama
# Restart=always
# RestartSec=3
# Environment="PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin"
#
# [Install]
# WantedBy=default.target
```

```
root@kali:~/ollama# systemctl status ollama
* ollama.service - Ollama Service
   Loaded: loaded (/etc/systemd/system/ollama.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/ollama.service.d
            └─override.conf
   Active: active (running) since Wed 2025-07-01 09:10:07 UTC; 14s ago
   Main PID: 2160 (ollama)
   Tasks: 6 (limit: 8000)
   Memory: 18.3M (peak: 31.7M)
   CPU: 336ms
   Group: system.slice/ollama.service
   CGroup: /system.slice/ollama.service
           └─2160 /usr/local/bin/ollama serve

Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.389Z level=INFO source=routes.go:1961 msg="server config" envs="map[CUDA_VISIBLE_DEVICES: GEMM_VA_VISIBLE_DEVICES: GPU_DEVICE_ORDINAL: HWP_VISIBLE_DEVICES: HSA_OVERRIDE_GFX_VERSION: HTTPS_PROXY: HTTP_PROXY:]"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.389Z level=INFO source=routes.go:1963 msg="ollama cloud disabled: false"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.389Z level=INFO source=images.go:883 msg="total blobs: 0"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.389Z level=INFO source=images.go:890 msg="total unused blobs removed: 0"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.389Z level=INFO source=routes.go:2023 msg="listening on {::}11484 (version 0.11.1)"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.390Z level=INFO source=model_list_cache.go:111 msg="model list cache hydration complete" models=0 failures=0 elapsed=169.79µs
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.392Z level=INFO source=runner.go:60 msg="discovering available GPUs: ."
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.409Z level=INFO source=types.go:50 msg="inference compute" idcpu.libraryncpu.compute"" namecpu.descriptioncpu.libdirsollama.driver"" pci.id"" type"" total="7.7 GiB" available="7.1 GiB"
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.409Z level=INFO source=routes.go:2073 msg="vram-based default context" total_vram="0 B" default_max_ctx=4096
Jul 01 09:10:07 techcorp-ollama-api ollama[2160]: time=2025-07-01T09:10:07.431Z level=INFO source=model_recommendation.go:177 msg="model recommendations cache sleep scheduled" wait=850m0.023512588s consecutive_failures=0
lines 1-22/22 (END)
```

3. Inference Engine Deployment (Ollama)

The Ollama engine was installed in a system service environment (systemd). A critical configuration change was made to expose the API on all network interfaces (variable OLLAMA_HOST=0.0.0.0), thereby allowing remote communication. A specific model, techcorp-financial, was compiled with strict parameters (temperature 0.1) to guarantee the accuracy of financial responses.

```
ollama create techcorp-financial -f ModelFile
gathering model components
pulling manifest
pulling b5370915da3 100%
pulling c806dc155a 100%
pulling c59f56a78a 100%
pulling 60ac4c62a77 100%
pulling 861187aa5b 100%
verifying sha256 digest
writing manifest
success
using existing layer sha256:b5370915da330c93d4f39f81b0d2c0b8a533f0f25e219576c956c268b6eb
using existing layer sha256:c806dc155a0303c36ab4a731a5a3c2d5c3d6c111e0d3c7b72b4
using existing layer sha256:c59f56a78a7d53b8eef37f0abb0a8a03871b2f63c311b9a024891da
creating new layer sha256:f0c13d0c47a5318cc70f09c0d0e12a5f40e7313ccacc22492b0386a090e0a
creating new layer sha256:edf023f0a5131a5a3c2d5c3d6c111e0d3c7b72b4
writing manifest
success
azureuser@techcorp-ollama-api:~$
```

4. Connectivity Test and Validation

The final validation of the infrastructure was performed by a remote REST request test. This step confirms that the server is correctly exposed on the Internet and that the financial model is ready to process JSON requests sent by the interfaces developed by the Web team.

```
azureuser@techcorp-ollama-api:~$ curl -X POST http://localhost:11434/api/generate -d '{
  "model": "techcorp-financial",
  "prompt": "What is the definition of ROI? Be very brief.",
  "stream": true
}'
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:47.457471598Z","response":"Return","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:47.85329499Z","response":" on","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:48.245449746Z","response":" In","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:48.637654748Z","response":"vest","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:49.029737783Z","response":"ment","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:49.430800063Z","response":" ("","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:49.833117082Z","response":"RO","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:50.230512243Z","response":"I","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:50.636207245Z","response":"),"done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:51.037984712Z","response":" measures","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:51.438908838Z","response":" the","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:51.83736052Z","response":" gain","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:52.237494723Z","response":" or","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:52.637476969Z","response":" loss","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:53.038617038Z","response":" generated","done":false}
{"model":"techcorp-financial","created_at":"2026-07-01T09:16:53.441695367Z","response":" on","done":false}
```

```
PS C:\Users\Black_Phanteks> Invoke-WebRequest -Uri "http://20.199.64.128:11434/api/generate" -Method Post -Body (ConvertTo-Json @({
  >> model = "techcorp-financial"
  >> prompt = "Hello"
  >> stream = $false
  >> }) -ContentType "application/json"

model          : techcorp-financial
created_at     : 2026-07-01T09:21:54.1010252Z
response       : Hello! I'm Phi, an AI language model here to assist you with financial inquiries or any questions related to economics that TechCorp Industries might have. How can I help today?
done           : True
done_reason    : stop
context        : {32086, 29871, 13, 13...}
total_duration : 15911716267
load_duration  : 78615841
prompt_eval_count : 63
prompt_eval_duration : 1592985800
eval_count     : 46
eval_duration  : 18889584900

PS C:\Users\Black_Phanteks>
```

HOW TO USE IT !!! DEVS !!!

API Documentation: TechCorp AI The inference infrastructure is deployed, secured, and operational. Please use the following parameters to integrate the `techcorp-financial` model into your web interface.

1. Endpoint

- **Base URL:** `http://20.199.64.120:11434`
- **Inference Endpoint:** `/api/generate`

2. Request Specifications (POST) Send a JSON object with the following parameters:

JSON

```
{  
  "model": "techcorp-financial",  
  "prompt": "YOUR_QUESTION_HERE",  
  "stream": true  
}
```

- **model:** Must be set to **techcorp-financial**.
- **stream:** Set to **true** to enable token-by-token streaming, providing a smoother user experience.

3. Integration Example (Python/Requests)

Python

```
import requests
```

```
url = "http://20.199.64.120:11434/api/generate"  
payload = {  
    "model": "techcorp-financial",  
    "prompt": "What are the Q3 financial forecasts?",  
    "stream": False  
}
```

```
response = requests.post(url, json=payload)  
print(response.json()['response'])
```

Notes for the Dev Team:

- The infrastructure is hosted on an isolated Azure VM. Network security rules have been configured to allow incoming traffic exclusively on port **11434**.
- The model is optimized with a temperature setting of **0.1** to ensure analytical precision and stable outputs.
- If you encounter connection timeouts, ensure your requests are targeting the correct public IP: **20.199.64.120**.